ADITYA MAHAMUNI

APPLIED AI ENGINEER

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PROFESSIONAL SUMMARY

Applied AI Engineer focused on evaluating LLMs and AI agents, with experience in synthetic-data pipelines, observability, regression analysis, and adversarial testing. Combines Python software engineering with cross-functional delivery to assess system behaviour and support reliable iteration in complex environments.

TECHNICAL EXPERTISE

Applied AI: Agentic AI, LLM Applications, Tool Use, RAG, Retrieval, Context/Prompt Engineering, Structured Outputs, MCP, Orchestration; LangGraph, OpenAI Agents SDK

Evaluation & ML: AI & LLM Evaluation, Synthetic Data Generation, NLP, Transformers, PyTorch, TensorFlow, Fine-Tuning and LoRA

Engineering & Platforms: Python, C++, SQL, FastAPI, Pydantic, PostgreSQL, Kafka, REST, JWT/RBAC, Docker, Linux, GitLab CI, Jenkins, AWS Bedrock / SageMaker, Git

EXPERIENCE

NatWest, London, UK | **Machine Learning Engineer** · *Dec 2024 - Present*

- Built workflows using observability tools and evaluation frameworks to assess LLM and agent outputs, retrieval, tool use, and failure modes, supporting reliable iteration and regression investigation.
- Built a synthetic-data pipeline for complex, multi-system workflows, enabling rapid evaluation experiments, model regression analysis, and adversarial testing without production data; reduced issue diagnosis and resolution turnaround time by 70%.
- Contributed to a production FastAPI backend microservice for the team's automated complaints investigation system, supporting AI-agent orchestration, execution tracking, persistence, and feedback APIs.
- Prototyped LangGraph, OpenAI Agents SDK, smolagents, and AutoGen; designed reusable routing, handoff, parallel, and evaluator-optimizer patterns and assessed production trade-offs.
- Built MCP tools and integrations for knowledge retrieval, developer platforms, and custom services to support multi-step agent workflows.
- Led cross-team delivery, owned technical design decisions, and communicated progress and outcomes to stakeholders and leadership.
- Contributed to a team project fine-tuning a language model using LoRA for COBOL code completion, infilling, and domain-specific question answering, including task evaluation and inference integration through AWS SageMaker endpoints.

NVIDIA Corporation, Pune, India | **Software Developer** · *Jul 2019 - Aug 2023*

- Developed a scalable Python framework and training scenarios for autonomous-vehicle simulation to support scenario-driven development and testing.
- Optimised Python, GitLab CI, and Docker delivery pipelines to improve deployment efficiency and accelerate scenario updates.
- Built Jenkins/Python automated testing frameworks that reduced manual testing effort and shortened test cycles.
- Developed ML tooling and CI/CD workflows using NLP, PyTorch, and TensorFlow to support development and experimentation.
- Created a Python/Linux diagnostic tool that reduced platform-specific issues and improved debugging reliability.

PROJECTS

Financial Event Extraction Using NLP | MSc Dissertation

Applied zero-shot and few-shot LLM approaches and fine-tuning to extract and classify financial events from news articles; compared generative modelling approaches for a financial-language task.

Financial Health and Stock Performance Analysis

Built Python-based financial analysis using clustering and regression to explore financial health and market performance. Original project reported 99% classification accuracy and regression RMSE of 1.118; task-specific methodology available in project materials.

Protein Expression Prediction

Developed image-based feature extraction using PCA, GLCM, and transfer learning; experimented with OLS, SVR, and CNN approaches to predict tissue protein expression, reporting CNN RMSE of 0.677.

EDUCATION

University of Warwick, UK | MSc Computer Science (2:1) | Sep 2023 - Sep 2024

Government College of Engineering, Aurangabad, India | BE Information Technology (7.448/10) | Jul 2015 - May 2019

CERTIFICATION

Neural Networks and Deep Learning - DeepLearning.AI | Sep 2020